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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 09/14/2010 |
| **PI Name:** | Strangman, Gary E Ph.D. | | |
| **Project Title:** | Objective Detection, Evaluation and Countermeasures for In-flight Depression | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Neurobehavioral and Psychosocial Factors Team | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **BHP**:Behavioral Health & Performance (archival in 2017) | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02129-2020 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2007 NSBRI-RFA-07-01 Human Health in Space |
| **Start Date:** | 10/01/2007 | **End Date:** | 09/30/2011 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Cartreine nee Carter, James ( Beth Israel Deaconess Medical Center )<br>Yeung, Albert ( Massachusetts General Hospital )<br>Fava, Maurizio ( Massachusetts General Hospital )<br>Feldman, Gregory ( Massachusetts General Hospital )<br>Zeffiro, Thomas ( Argosy Omnimedia, Inc. )<br>Zhang, Quan ( Massachusetts General Hospital ) | | |
| **Grant/Contract No.:** | NCC 9-58-NBPF01301 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Psychological stresses, including depression, can significantly disrupt one's ability to function effectively and efficiently, and the associated performance deficits can seriously jeopardize space mission success. The incidence of serious depression in Earth based analogues of the spaceflight environment has been reported as up to 13% per person per year. Extrapolating from existing reports of depressive episodes during short-duration spaceflight, depression is thus a probable condition in one or more members of a five to seven person crew during a long duration spaceflight (e.g., a 30 month mission to Mars). Mission success can be jeopardized by depression either directly, from the potentially life threatening consequences of lapses in performance, or indirectly, by adding to the workload and stress of other crewmembers. The likelihood and potentially serious consequences of depression during spaceflight explains why human performance failure due to mood alterations such as depression, anxiety, or other psychiatric and cognitive problems is a high priority risk for all mission types (ISS, Moon, Mars). Countermeasures are already in place: medications and psychological consultations with ground-crews. However, current in-flight methods to decide whether a countermeasure should be used rely heavily on subjective self-reports. The biological basis of mood disorders suggests that neural biomarkers may provide a more objective method for assessing depression. Aim 1 of this proposal, therefore, seeks to identify neural biomarkers sensitive to, and specific for, depression. These measures will be used in evaluating and validating a flight-capable, noninvasive neuroimaging technology (near-infrared neuroimaging, or NIN) for its ability to detect biomarkers of depression and its severity. In Aim 2, we seek to develop and improve hardware and software for mobile neuroimaging, including the development of ultra lightweight NIN instruments. For Aim 3, we will focus on evaluating the novel hardware for detection of the biomarkers identified in Aim 1. Together, successful completion of these aims will provide a new, mobile neuroimaging technology suitable for detecting biomarkers associated with depression. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Successful completion of this project would lead to the following Earth-relevant benefits:<br>1. Depression Biomarkers: We seek to identify biomarkers that are suitable for (i) depression diagnosis, and (ii) assessing depression severity. If a reliable NIN-based biomarker is identified, this would provide initial validation for NIN-based evaluation of depressed individuals. If such brain-based biomarkers can be reliably measured in an office setting, it would provide more objective tracking of depression severity and potentially helping to optimize delivery of all types of depression therapies. A NIN system for depression assessment has the further potential to be relatively low cost, thereby facilitating access to such capabilities in broader regions than currently possible, including rural and otherwise underserved communities.<br>2. Mobile Neuroimaging: The technologies we are developing are designed to enable neuroimaging in mobile environments, including spaceflight analogs and spaceflight itself. Such technologies have the potential to impact a wide range of novel brain monitoring applications on Earth as well. Such domains range from mobile epilepsy monitoring, to monitoring treatment efficacy via brain imaging during an office visit, to first-responder or battlefield head trauma evaluations, as well as generally more available, less expensive methods for diagnosing, monitoring, and treating not only depression, but other psychiatric or cognitive disorders involving alterations in brain function. NIN is of particular promise as a brain imaging technology as it is sufficiently low-cost, robust and portable to be made readily available in diverse operational environments including urban, rural, and remote settings. |
| **Task Progress:** | In the past year we have made significant progress on three primary fronts: experimental, computational, and device development. Experimentally, having previously completed development of optimized neuroimaging protocols and tasks, we continued neuroimaging data collection for the main depression study. Initial results suggest that both our MRI and near-infrared neuroimaging (NIN) measures are quite sensitive to task-related brain activity alterations in depressed participants. The tasks we have implemented will allow us to examine complex sensorimotor performance in much finer detail than previously investigated. We also completed data collection and preliminary analysis for a laboratory test of NIN under head-down tilt (HDT) conditions. This study was encouraging, as NIN was found to be only slightly less sensitive to brain function during cephalad fluid shifts associated with both HDT and spaceflight.<br>On the computational front, in the last year we completed an additional 20,000 hours of computer processing time to generate even more detailed models of the distribution of photons migrating through the head from each of 3,555 separate injection points (5mm spacing) around the entire scalp. The simulation data provide the most detailed maps of brain sensitivity available for NIN measurements. A manuscript describing the results is nearing completion, and software to use these results for NIN data processing and imaging is under development.<br><br>In the last year, we have also made considerable progress in terms of device development, achieving four major milestones regarding lightweight, mobile brain monitoring devices. First, we optimized the performance of our portable near-infrared neuroimaging (NIN) instrument NINscan 2a (previously named OpticHolter 2a). This made the device suitable for strap-and-go measurements on a wide range of locations on the head. Second, in January 2010 and again in August 2010, we successfully tested NINscan 2a for operational performance at high altitude via testing at various stages during separate hikes up Mt. Kilimanjaro. Third, in March 2010 we completed a third generation prototype device, NINscan 3a. This device includes multiple enhancements over version 2a, including multi-device synchronization capabilities, substantial noise reduction, 50% improvement in battery life, and the potential to use dual-color laser diodes in place of LED light sources. Finally, we completed a third distinct prototype device, NINscan TD. This system has exceptional noise characteristics, and the light sources and detectors in NINscanTD are fully controlled by the embedded microcontroller, thereby enabling the use of time division (TD) multiplexing. The TD approach was an essential milestone prior to scaling up the device for broader spatial neuroimaging coverage, with the goal of whole-head neuroimaging. Together, these efforts strongly support the goal of developing technologies suitable for brain imaging in spaceflight. |
| **Bibliography Type:** | Description: (Last Updated: 07/08/2026) |
| **Abstracts for Journals and Proceedings** | Strangman GE, Supelana C, Holper L, Zeffiro TA. "Brain activity changes measured with near-infrared neuroimaging in a head-down tilt spaceflight analog." 81st Annual Scientific Meeting of the Aerospace Medical Association, Phoenix, AZ, May 9-14, 2010.<br>Aviation, Space, and Environmental Medicine 2010 Mar;81(3):263. , Mar-2010 |

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| **Abstracts for Journals and Proceedings** | Zeffiro TA, Zhang Q, Strangman GE. "Brain hemodynamic changes measures with near-infrared spectroscopy during altered gravity." Next-Generation Suborbital Researchers Conference, Boulder, CO, February 18-20, 2010. Next-Generation Suborbital Researchers Conference Program, February 2010. Abstract #4016. http://www.lpi.usra.edu/meetings/nsrc2010/pdf/4016.pdf , Feb-2010 |
| **Abstracts for Journals and Proceedings** | Zeffiro TA, Zhang Q, Supelana C, Strangman GE. "Brain hemodynamic changes in microgravity measured with near-infrared neuroimaging." 81st Annual Scientific Meeting of the Aerospace Medical Association, Phoenix, AZ, May 9-14, 2010. Aviation, Space, and Environmental Medicine 2010 Mar;81(3):263. , Mar-2010 |